Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/11/2008
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Development of a User Interface for the PVT SelfTest (PST)		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	09/01/2007	**End Date:**	08/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-NBPF00002		
Performance Goal No.:			
Performance Goal Text:			

Space operations require that astronauts maintain the highest level of performance capability throughout their missions to mitigate risk in what is a high-risk operational environment. A brief sensitive objective self-test of performance fitness for duty is needed to inform astronauts of their performance fitness and need for countermeasures.
The Principal Investigator developed a Psychomotor Vigilance Test (PVT) that has been validated to detect cognitive deficits caused by a variety of factors in space flight (e.g., sleep loss, sleep shifts, motion sickness). The PVT is optimized for repeated use, because it is free of learning effects and aptitude differences that plague other cognitive measures. The Principal Investigator used the 10-min PVT to develop a specially programmed 3-minute PVT SelfTest, and validated it to be as sensitive as the 10-min PVT to total sleep loss, based on optimization algorithms. The 3-min. PVT SelfTest has been deployed in three NASA Extreme Environment Mission Operations (NEEMO) missions (9, 12, and 13) and found to be acceptable to astronauts and sensitive to their activity demands. However, the PVT SelfTest use

Task Description:	in NEEMO did not have an interface that informed astronauts of their performance relative to their own behavioral norms. This project seeks to develop the PVT SelfTest user interface to immediately inform astronauts (post test) of their performance fitness and need for countermeasures. The project has 4 specific aims: (1) Use PVT SelfTest data acquired in NEEMO 9, NEEMO 12, and NEEMO 13 to develop astronaut performance norms, and to compare this data to other experimental and clinical research norms from large scale studies the PI has performed for NIH, AFOSR (Air Force Office of Scientific Research), and NASA to set parameter limits for information feedback via the user interface; (2) Create quality control parameters to certify a PVT SelfTest trial is valid; (3) Determine if different options are required for data interpretive display of performance on the PVT SelfTest; and (4) Computer program of the results of Aims 1-3 into the current Windows PVT SelfTest architecture. The result of project works is a performance feedback interface to augment the brief, validated, zero upmass, performance test to provide astronauts with immediate feedback about cognitive deficits by displaying information about impulsivity, lapsing of attention, reaction speed, overall performance capability, and test compliance. The PVT SelfTest has the potential to yield an objective, practical method to inform astronauts about performance changes to aid in the selection of appropriate fatigue countermeasures. The PVT SelfTest with the performance feedback interface developed in this project has been approved for a flight experiment beginning with International Space Station (ISS) Increment 21-22.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary aim of this project was to develop a brief, validated, performance feedback interface to provide astronauts with immediate feedback about cognitive deficits caused by sleep loss, sleep shifts, medication use, and other factors that may degrade performance. These factors are present during Earth-based operations by astronauts. The PVT SelfTest may provide astronauts with a useful performance monitoring tool to aid in the selection of fatigue countermeasures during training and mission preparation; especially when operations include sleep loss, night work, and travel to Russia. Additionally, the PVT SelfTest can be adapted to provide mission control personnel with a tool to provide individualized performance capability feedback during continuous mission support operations by establishing appropriate normative data representative of mission controllers.
Task Progress:	The overarching goal of the project is to provide a brief, validated, zero upmass, performance test to provide astronauts with immediate feedback about cognitive deficits caused by a variety of factors in space flight (e.g., sleep loss, sleep shifts, medication use). Substantial progress has been made on developing astronaut norms based on N=241 PVT SelfTest trials acquired on 11 astronauts (aquanauts) and other crew acquired in NEEMO 9, NEEMO 12, and NEEMO 13. Significant advances were made in the development of an algorithm that calculates a numerical performance index to provide astronauts with feedback about their PVT SelfTest performance that is both informative about the test performance and about the validity of the test. Prototype data interpretation displays were generated that varied both the type of performance information displayed and the method of display (e.g., numerical, graphical). The performance feedback algorithm and display were programmed into the current Windows PVT SelfTest architecture by Pulsar Informatics Inc. A demonstration version of full PVT SelfTest including performance feedback algorithm and display was generated for evaluation by NEEMO astronauts for refinement and preparation for validation on ISS.
Bibliography Type:	Description: (Last Updated: 06/30/2026)